

大模型RAG进阶实战营 – 准备篇

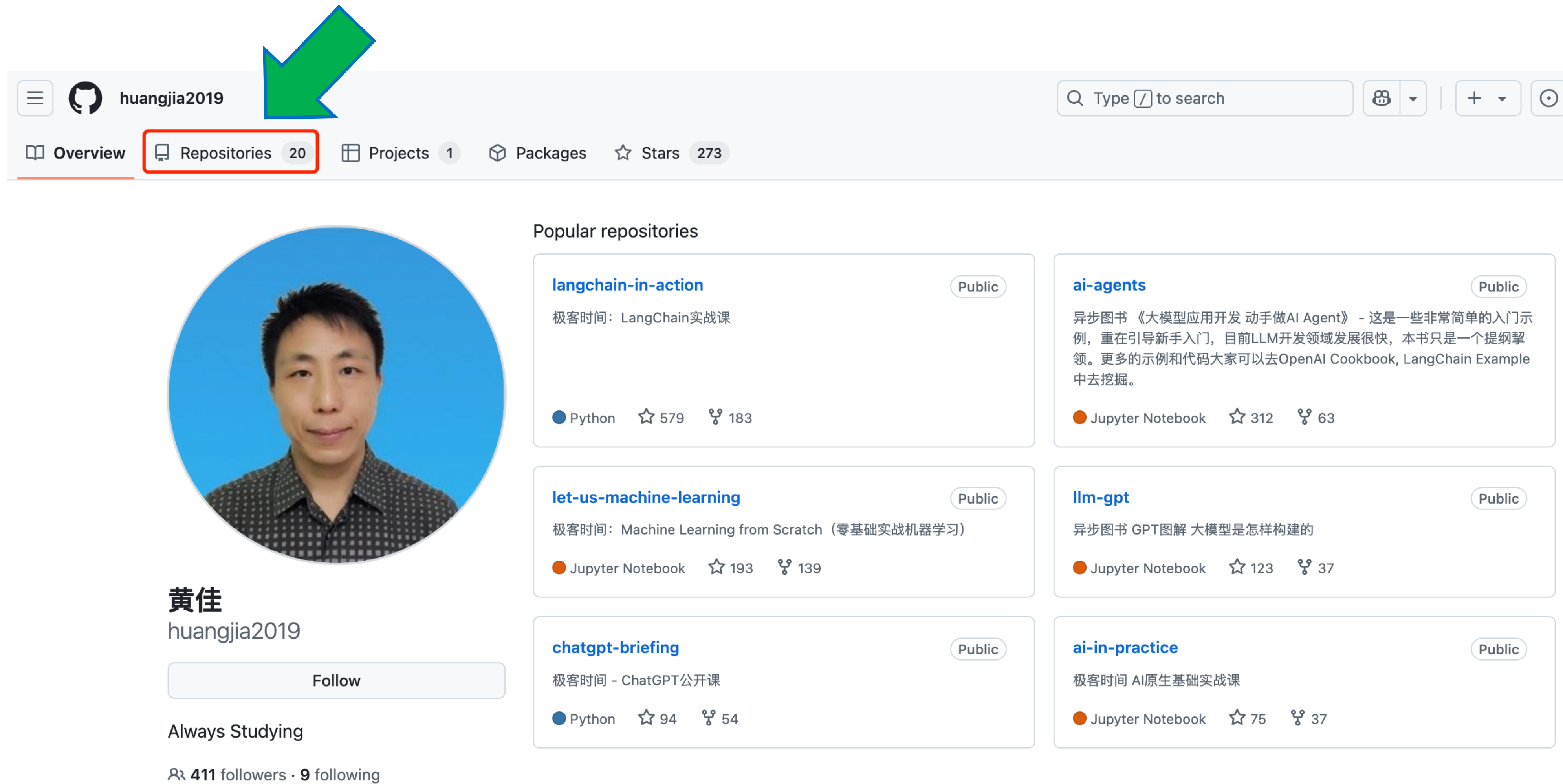
代码下载及安装系统环境

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访问 <https://github.com/huangjia2019> , 然后点击“**Repositories**”

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极客时间RAG训练营, RAG 10大组件全面拆解, 4个实操项目吃透 RAG 全流程。

agentai ragllm

Python 5 3 Updated 2 days ago

rag-project03-auditPublic

ESG合规审计系统 - 可持续发展报告检查工具

Python 4 Updated 4 days ago

点击进入名为“rag-in-action”这个仓库

https://github.com/huangjia2019/rag-in-action

 rag-in-action Public

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master 1 Branch 0 Tags

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Code

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3231cf3 · 2 days ago 15 Commits

00-简单RAG-SimpleRAG	祝大家学习快乐	2 days ago
90-文档-Data	祝大家学习愉快	3 days ago
91-环境-Environment	祝大家学习愉快	2 days ago
92-图片-Pic	祝大家学习愉快	3 days ago
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About

极客时间RAG训练营, RAG 10大组件全面拆解, 4个实操项目吃透 RAG 全流程。

u.geekbang.org/subject/airag/1009927

agent

ai

rag

llm

Readme

Activity

5 stars

3 watching

3 forks

Report repository

使用Git命令“**git clone** <https://github.com/huangjia2019/rag-in-action.git>” 将仓库代码克隆至本地

<https://github.com/huangjia2019/rag-in-action>

 **rag-in-action** Public

Watch 3 Fork 3 Star 5

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Local Codespaces

Clone ?

HTTPS SSH GitHub CLI

https://github.com/huangjia2019/rag-in-act: 

Clone using the web URL.

Open with GitHub Desktop

Download ZIP

About

极客时间RAG训练营, RAG 10大组件全面拆解, 4个实操项目吃透 RAG 全流程。

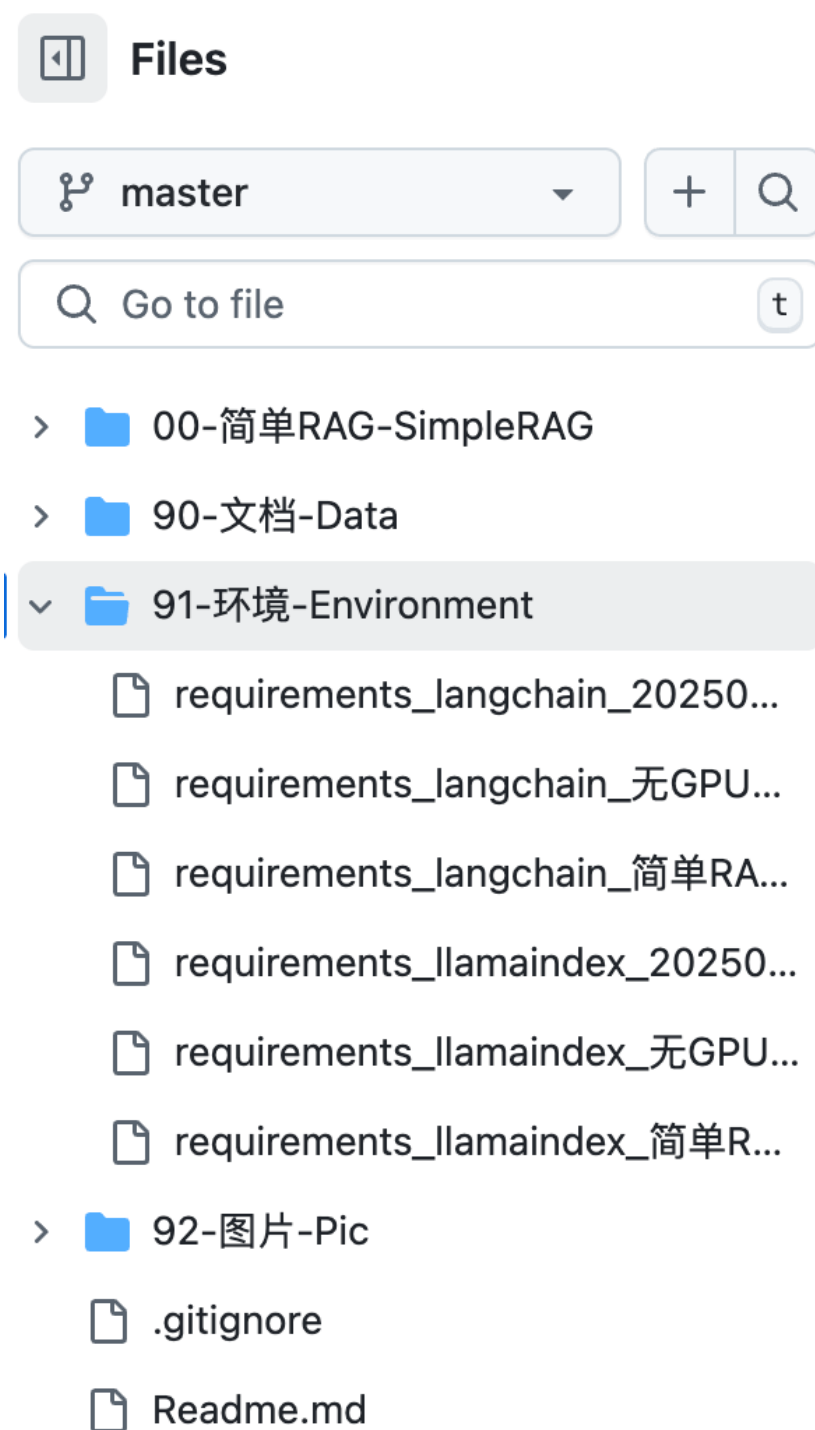
u.geekbang.org/subject/airag/1009927

agent ai rag llm

Readme Activity 5 stars 3 watching 3 forks Report repository

如果对**Git**命令不熟悉，可以点击绿色的“**Code**”下拉按钮，选择“**Download ZIP**”，将打包后代码文件下载至本地

安装Python环境



rag-in-action / 91-环境-Environment /

Add file ...

huangjia2019 祝大家学习愉快 3231cf3 · 2 days ago History

Name	Last commit message	Last commit date
..		
requirements_langchain_20250305.txt	祝大家学习愉快	3 weeks ago
requirements_langchain_无GPU版.txt	祝大家学习愉快	4 days ago
requirements_langchain_简单RAG.txt	祝大家学习愉快	2 days ago
requirements_llamaindex_20250305.txt	祝大家学习愉快	3 weeks ago
requirements_llamaindex_无GPU版.txt	祝大家学习愉快	4 days ago
requirements_llamaindex_简单RAG.txt	祝大家学习愉快	2 days ago

这里是我们项目要用到的Python包，建议先使用红框框起来的两个文件来批量安装项目所需的包。

创建对应的Python虚拟开发环境

`python3 -m venv 20250306_LangChain` （红色部分可以换成自己喜欢的名字）

`python3 -m venv 20250306_LlamaIndex` （红色部分可以换成自己喜欢的名字）

```
huangj2@IHP-SPD-CF00015: /  
~$ cd /mnt/external_disk/venv  
/mnt/external_disk/venv$ ls  
50203_LangChain  20250203_SimplePython  20250214_RAG_Framework  backup_requirements.sh  
/mnt/external_disk/venv$ python3 -m venv 20250306_LangChain  
/mnt/external_disk/venv$ ls  
50203_SimplePython  20250306_LangChain  
50214_RAG_Framework  backup_requirements.sh  
/mnt/external_disk/venv$ |
```

新创建的环境

激活这个新环境

Linux、macOS 等系统使用如下命令激活

source 20250306_LangChain/bin/activate

Windows 系统使用如下命令激活

20250306_LangChain\Scripts\activate

```
(20250306_LangChain) :/mnt/external_disk/venv$ source 20250306_LangChain/bin/activate  
/mnt/external_disk/venv$ |
```



更新 pip 版本

```
pip install --upgrade pip
```

通过 requirements 文件批量安装对应项目环境所需要的包

```
pip install -r /你的目录/requirements_langchain_简单RAG.txt
```

Files

master

Go to file

> 00-简单RAG-SimpleRAG

> 90-文档-Data

> 91-环境-Environment

> 92-图片-Pic

.gitignore

Readme.md

rag-in-action / 91-环境-Environment /

Add file

...

huangjia2019 祝大家学习愉快

3231cf3 · 2 days ago

History

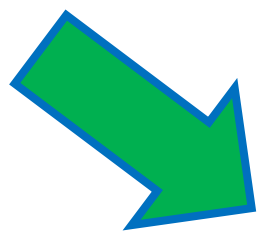
Name	Last commit message	Last commit date
..		
requirements_langchain_20250305.txt	祝大家学习愉快	3 weeks ago
requirements_langchain_无GPU版.txt	祝大家学习愉快	4 days ago
requirements_langchain_简单RAG.txt	祝大家学习愉快	2 days ago
requirements_llamaindex_20250305.txt	祝大家学习愉快	3 weeks ago
requirements_llamaindex_无GPU版.txt	祝大家学习愉快	4 days ago
requirements_llamaindex_简单RAG.txt	祝大家学习愉快	2 days ago

pip check

```
(20250306_LangChain) [redacted]:/mnt/external_disk/venv$ pip check  
No broken requirements found.
```

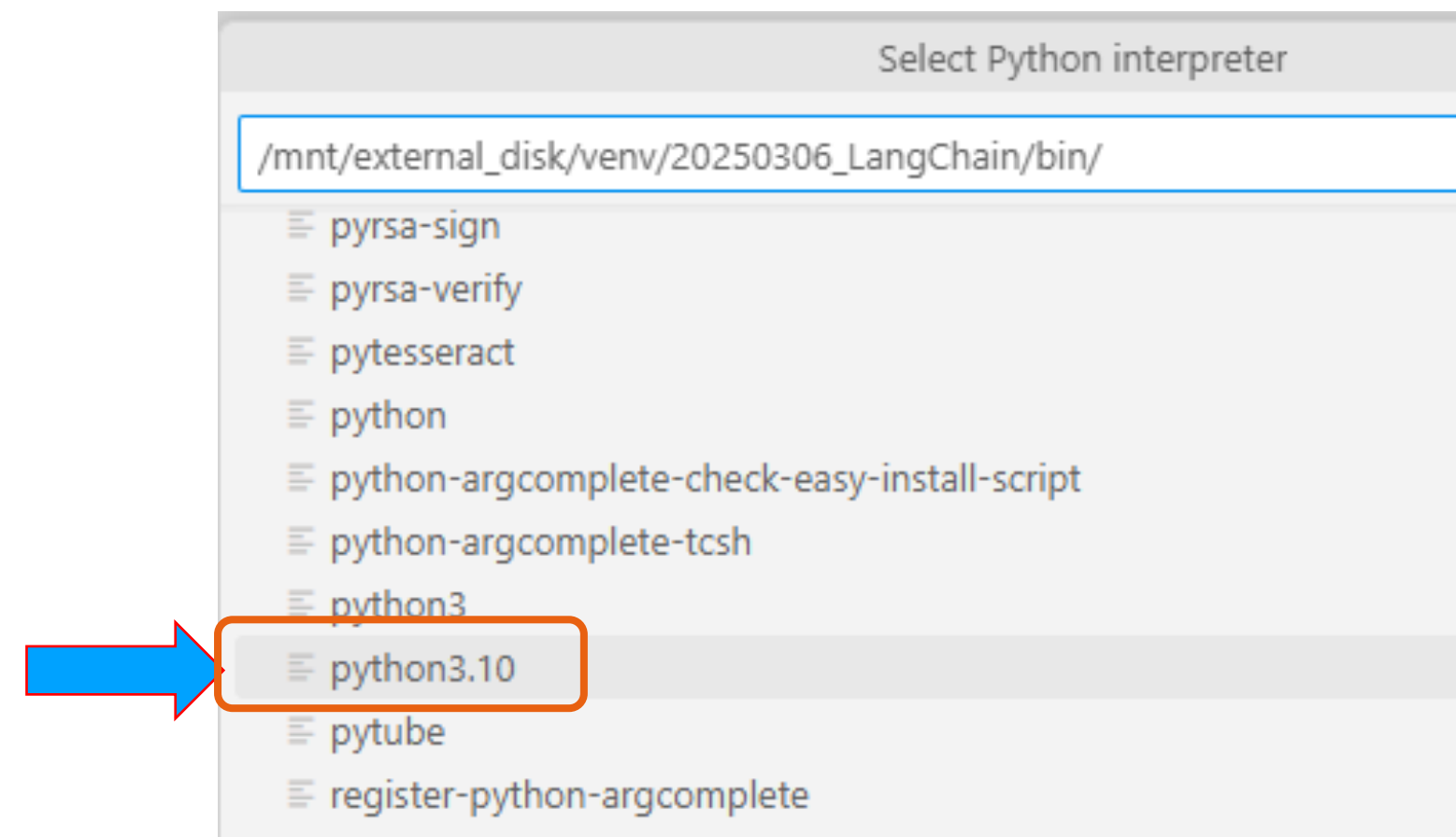
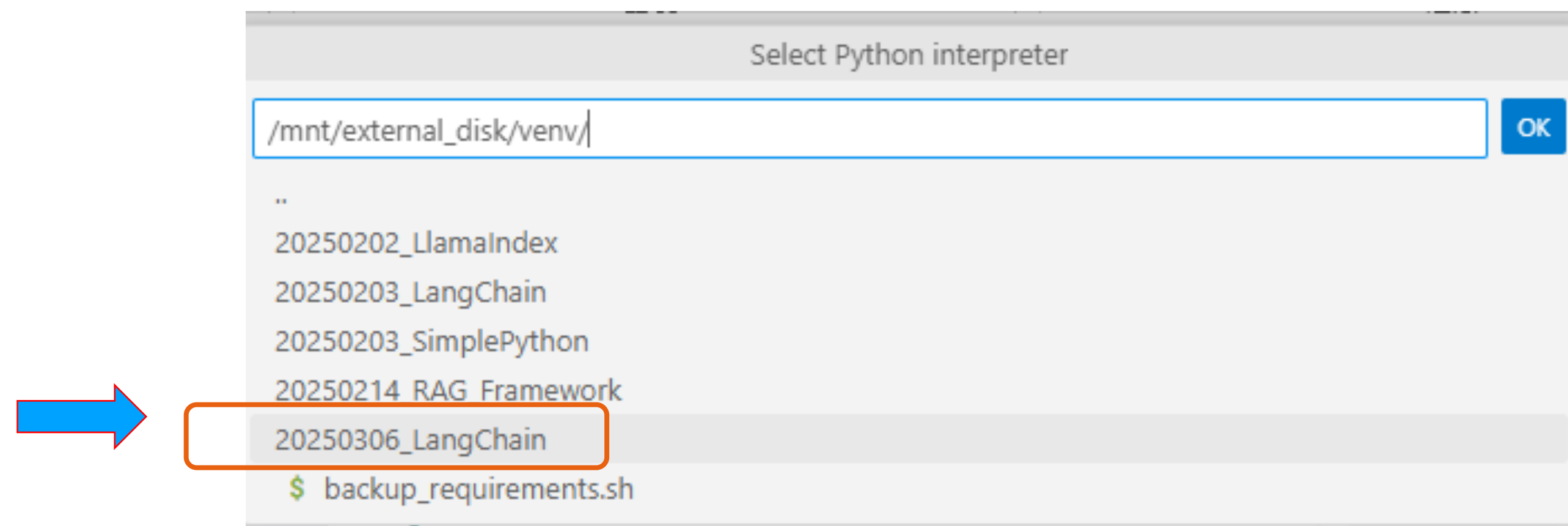
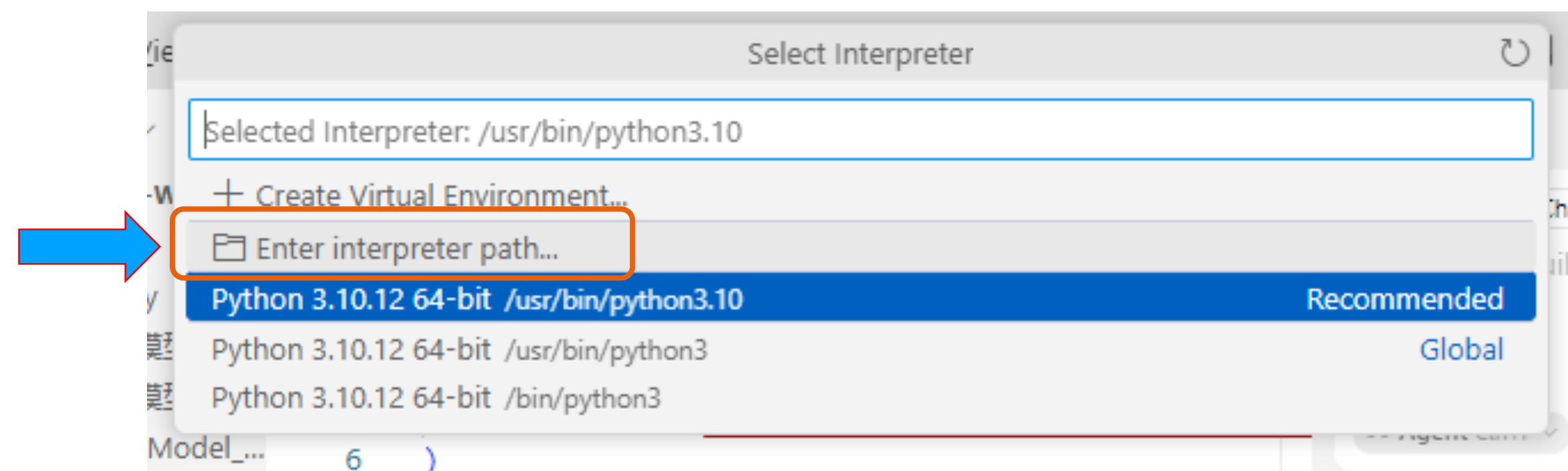
退出当前环境，重复之前的步骤激活下一个环境，并用对应的 **requirements** 文件安装对应的包

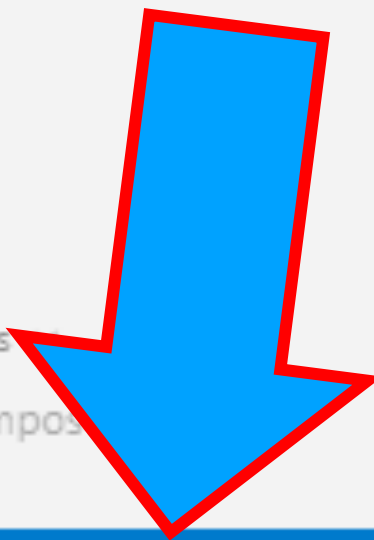
deactivate



```
(20250306_LangChain) [redacted]:/mnt/external_disk/venv$ deactivate  
[redacted]/mnt/external_disk/venv$
```

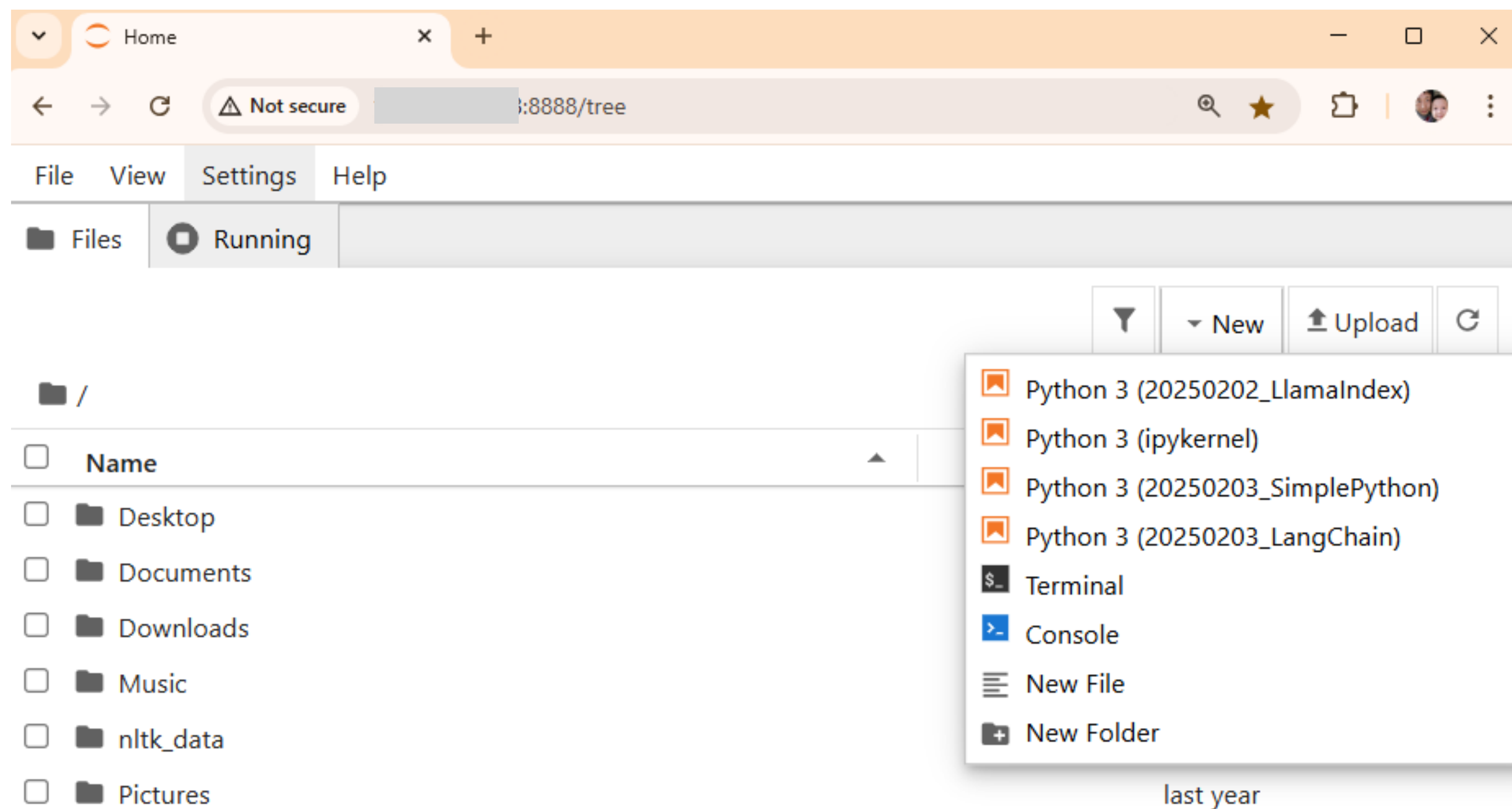






选好之后的效果

安装并配置Jupyter Notebook



安装 Jupyter Notebook

安装 Jupyter Notebook

```
pip install notebook
```

激活环境

```
source 20250306_LangChain/bin/activate
```

安装 ipykernel

```
pip install ipykernel
```

将该环境注册到 Jupyter

```
python -m ipykernel install --user --name 20250306_LangChain --display-name "Python 3 (20250306_LangChain)"
```

退出环境

```
deactivate
```

jupyter notebook

运行Jupyter Notebook

03_LangGraph_RAG Last Checkpoint: 3 hours ago

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (20250203_LangChain)

[11]: # 1. 加载文档

```
from langchain_community.document_loaders import WebBaseLoader
loader = WebBaseLoader(
    web_paths=("https://zh.wikipedia.org/wiki/%E9%BB%91%E7%A5%9E%E8%AF%9D%E7%BC%9A%E6%82%9F%E7%A9%BA",)
)
docs = loader.load()
```

[2]: # 2. 文档分块

```
from langchain_text_splitters import RecursiveCharacterTextSplitter
text_splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=200)
all_splits = text_splitter.split_documents(docs)
```

[3]: # 3. 设置嵌入模型

```
from langchain_huggingface import HuggingFaceEmbeddings
embeddings = HuggingFaceEmbeddings(
    model_name="BAAI/bge-small-zh-v1.5",
    model_kwargs={'device': 'cpu'},
    encode_kwargs={'normalize_embeddings': True}
)
```

/mnt/external_disk/venv/20250203_LangChain/lib/python3.10/site-packages/tqdm/auto.py:21: TqdmWarning: IPProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
from .autonotebook import tqdm as notebook_tqdm

[4]: # 4. 创建向量存储

```
from langchain_core.vectorstores import InMemoryVectorStore
vector_store = InMemoryVectorStore(embeddings)
vector_store.add_documents(all_splits)
```

选择匹配的环境

jupyter kernelspec list

确认 Jupyter 内核是否正确安装

```
:~$ jupyter kernelspec list
Available kernels:
  20250202_llamaindex      /home/huangj2/.local/share/jupyter/kernels/20250202_llamaindex
  20250203_langchain       /home/huangj2/.local/share/jupyter/kernels/20250203_langchain
  20250203_simplepython    /home/huangj2/.local/share/jupyter/kernels/20250203_simplepython
  python3                  /home/huangj2/.local/share/jupyter/kernels/python3
```

使用 Colab



00_简单RAG-LangChain.ipynb ☆ ☁

文件 修改 视图 插入 代码执行程序 工具 帮助

<https://colab.research.google.com/>

🔍 命令

+ 代码 + 文本



```
# ### 设置大模型地址和API KEY
# import os
# os.environ['OPENAI_API_KEY'] = 'XXX'
# os.environ['OPENAI_BASE_URL'] = 'https://vip.apiyi.com/v1'

# # 设置环境变量 DEEPSEEK_API_KEY 的值为 "your_api_key_value"
# os.environ['DEEPSEEK_API_KEY'] = 'XXX'

### 设置大模型地址和API KEY https://platform.openai.com/api-keys
import os
os.environ['OPENAI_API_KEY'] = 'sk-BUovI8yXVWxvv4ErF0F76e0968De486e82B9F0338264Aa47'
os.environ['OPENAI_BASE_URL'] = 'https://vip.apiyi.com/v1'
# !export OPENAI_API_KEY=sk-paSHgQoVeKaglrrou9d81Fa2f534940C1Ba394f02C45aF3D2
# !export OPENAI_BASE_URL=https://vip.apiyi.com/v1

# # 设置环境变量 DEEPSEEK_API_KEY 的值为 "your_api_key_value"
os.environ['DEEPSEEK_API_KEY'] = 'sk-8581a44ae63942ed83872c72a3b6f482'
```

▼ 实战1: 02_LangChain_DeepSeek_Model

```
# 1. 加载文档
import os
from dotenv import load_dotenv

# 加载环境变量
load_dotenv()

from langchain_community.document_loaders import WebBaseLoader

loader = WebBaseLoader(
    web_paths=("https://zh.wikipedia.org/wiki/%E9%BB%91%E7%A5%9E%E8%AF%9D%EF%BC%9A%E6%82%9F%E7%A9%BA",)
)
docs = loader.load()
```

设置大模型 API

deepseek 开放平台

用量信息

API keys

充值

账单

充值

<https://platform.deepseek.com/>

USD CNY

支付金额

¥10 ¥20 **¥50** ¥100 ¥300 ¥500 自定义 查看价格

【错峰优惠活动】北京时间每日 00:30-08:30 为错峰时段，API 调用价格大幅下调：DeepSeek-V3 降至原价的 50%，DeepSeek-R1 降至 25%，在该时段调用享受更经济更流畅的服务体验。【查看价格详情】

支付方式

支付宝

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去支付

提示

1. 充值金额仅用于调用 API 服务，网页版及 App 对话免费使用，无需充值。

2. 充值后可以前往账单页面开具发票。

将获取到的 **API Key** 设置到系统环境变量，具体步骤请参考：

<https://zhuanlan.zhihu.com/p/627665725>

DeepSeek API Key 的设置方法与此类似。

THANKS

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